

Risperidone

Therapeutics

Brands

- Risperdal
- Consta
- Perseris
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine, serotonin, norepinephrine receptor antagonist (DSN-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia, ages 13–17 (oral) and adults (oral, Consta, Perseris)**

- **Delaying relapse in schizophrenia (oral)**
- **Other psychotic disorders (oral)**
- **Acute mania/mixed mania, ages 10 and older (oral, monotherapy and adjunct to lithium or valproate)**
- **Autism-related irritability in children ages 5–16 (oral)**
- **Bipolar maintenance (oral, Consta, monotherapy and adjunct to lithium or valproate)**
- Bipolar depression
- Treatment-resistant depression
- Behavioral disturbances in dementias
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- Interactions at a myriad of other neurotransmitter receptors may contribute to risperidone's efficacy

* Specifically, 5HT₇ antagonist properties may contribute to antidepressant actions

How Long Until It Works

- Psychotic and manic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition and affective stabilization
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenic patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and

sustain long-term relationships

- Many bipolar patients may experience a reduction of symptoms by half or more
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot

injection

- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse



Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Other mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient

- is overweight (BMI 25.0–29.9)
- is obese (BMI >30)
- has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
- has diabetes (fasting plasma glucose >126 mg/dL)
- has hypertension (BP >140/90 mmHg)
- has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and risperidone should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia, especially at high

doses

- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

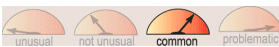
- * May increase risk for diabetes and dyslipidemia
- * Dose-dependent drug-induced parkinsonism
- * Dose-related hyperprolactinemia
- * Dose-dependent dizziness, insomnia, anxiety, sedation
- Nausea, constipation, abdominal pain, weight gain
- Tachycardia, dose-dependent sexual dysfunction
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- Rare orthostatic hypotension, usually during initial dose titration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures

Weight Gain



- Many experience and/or can be significant in amount
- Can become a health problem in some
- May be less than for some antipsychotics, more than for others

Sedation



- Many experience and/or can be significant in amount
- Usually transient
- May be less than for some antipsychotics, more than for others

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take at bedtime to help reduce daytime sedation
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Oral: 2–8 mg/day for acute psychosis and bipolar disorder
- Oral: 0.5–2.0 mg/day for children and elderly
- Consta: 12.5–50 mg every 2 weeks (deep intramuscular) (see section after Pearls for dosing and use)
- Perseris: 90 or 120 mg every month (subcutaneous) (see section after Pearls for dosing and use)

Dosage Forms

- Tablet 0.25 mg, 0.5 mg, 1 mg, 2 mg, 3 mg, 4 mg, 6 mg
- Orally disintegrating tablet 0.5 mg, 1 mg, 2 mg, 3 mg, 4 mg
- Liquid 1 mg/mL –30 mL bottle
- Risperidone long-acting depot microspheres formulation for deep intramuscular administration 12.5 mg vial/kit, 25 mg vial/kit, 37.5 mg vial/kit, 50 mg vial/kit
- Long-acting monthly formulation 90 mg, 120 mg

How to Dose –Oral

- In adults with psychosis in nonemergent settings, initial dosage recommendation is 1 mg/day orally in 2 divided doses

- Increase each day by 1 mg/day orally until desired efficacy is reached
- Maximum generally 16 mg/day orally
- Typically maximum effect is seen at 4–8 mg/day orally
- See also the Switching section, after Pearls
- Can be administered on a once daily schedule as well as twice daily orally

Dosing Tips –Oral

- * **Less may be more:** lowering the dose in some patients with stable efficacy but side effects may reduce side effects without loss of efficacy, especially for doses over 6 mg/day orally
- * Target doses for best efficacy/best tolerability in many adults with psychosis or bipolar disorder may be 2–6 mg/day (average 4.5 mg/day) orally
- Low doses may not be adequate in difficult patients
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine

- Approved for use up to 16 mg/day orally, but data suggest that risk of drug-induced parkinsonism is increased above 6 mg/day
- Risperidone oral solution is not compatible with cola or tea
- Children and elderly may need to have oral twice daily dosing during initiation and titration of drug dosing and then can switch to oral once daily when maintenance dose is reached
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- For treatment resistance, consider obtaining plasma drug levels to guide dosing above recommended levels
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Rarely lethal in monotherapy overdose; sedation, rapid heartbeat, convulsions, low blood pressure, difficulty breathing

Long-Term Use

- Approved to delay relapse in long-term treatment of schizophrenia
- Often used for long-term maintenance in bipolar disorder and various behavioral disorders
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- See Switching sections of individual agents for how to stop risperidone
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Metabolites are active
- Metabolized by CYP450 2D6
- Parent drug of oral formulation has 20–24 hour half-life
- Consta (long-acting intramuscular) has 3–6 day half-life and elimination phase of approximately 7–8 weeks after last injection
- Food does not affect absorption



Drug Interactions

- May increase effect of antihypertensive agents
- May antagonize levodopa, dopamine agonists
- Clearance of risperidone may be reduced and thus plasma levels increased by clozapine; dosing adjustment usually not necessary

- Coadministration with carbamazepine may decrease plasma levels of risperidone
- Coadministration with fluoxetine and paroxetine may increase plasma levels of risperidone
- Since risperidone is metabolized by CYP450 2D6, any agent that inhibits this enzyme could theoretically raise risperidone plasma levels; however, dose reduction of risperidone is usually not necessary when such combinations are used
- Perseris: if a strong CYP450 2D6 inhibitor is to be initiated, patients may be placed on the lowest dose (90 mg) 2–4 weeks before the start of the 2D6 inhibitor, to adjust for the expected increase in risperidone plasma concentrations
- Perseris: if a strong CYP450 3A4 inducer is to be initiated, patient should be closely monitored during the first 4–8 weeks; upward dose adjustment (for patients taking 90 mg) or oral supplementation (for patients taking 120 mg) may be necessary

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Risperidone should be used cautiously in patients at risk for aspiration pneumonia, as dysphagia has been reported
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation

- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and risperidone should be used cautiously in patients at risk for aspiration pneumonia
- Priapism has been reported
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If there is a proven allergy to risperidone or paliperidone

Special Populations

Renal Impairment

- Oral: initial 0.5 mg twice a day for first week; increase to 1 mg twice a day during second week; dosage increases above 1.5 mg twice a day should occur at least 1 week apart
- Consta: patient should demonstrate tolerability of at least 2 mg/day orally prior to initiation; should be dosed at 25 mg every 2 weeks; oral administration should be continued for 3 weeks after the first injection
- Perseris: has not been studied in this population; recommended to titrate patients to at least 3 mg/day of oral risperidone and establish efficacy/tolerability prior to initiating the monthly injection

Hepatic Impairment

- Oral: initial 0.5 mg twice a day for first week; increase to 1 mg twice a day during second week
- Consta: patient should demonstrate tolerability of at least 2 mg/day orally prior to initiation; should be dosed at 25 mg every 2 weeks; oral administration should be continued for 3 weeks after the first injection
- Perseris: has not been studied in this population; recommended to titrate patients to at least 3 mg/day of oral risperidone and establish efficacy/tolerability prior to initiating the monthly injection

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension

- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist
- * When administered to elderly patients with atrial fibrillation, may increase the chances of stroke

Elderly

- Initial 0.5 mg orally twice a day; increase by 0.5 mg twice a day; titrate once a week for doses above 1.5 mg twice a day
- Consta: recommended dose is 25 mg every 2 weeks; oral administration should be continued for 3 weeks after the first injection
- Perseris: not studied in this population
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

* Oral risperidone is approved for use in schizophrenia (ages 13 and older), mania/mixed episodes (ages 10 and older), and irritability associated with autism (ages 5–16)

* Risperidone is the most frequently used atypical antipsychotic in children and adolescents

- Clinical experience and early data suggest risperidone is safe and effective for behavioral disturbances in children and adolescents
- Children and adolescents using risperidone may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding

- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Early findings of infants exposed to risperidone in utero do not show adverse consequences
- When administered to pregnant rats and rabbits, risperidone was not teratogenic at doses up to 6 times the maximum recommended human dose (MRHD); increased stillbirths and decreased birth weight occurred at 1.5 times the MRHD when administered to pregnant rats
- Risperidone may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- Effects of hyperprolactinemia on the fetus are unknown
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypical/antipsychotic>

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on risperidone should be monitored for possible adverse effects; sedation, failure to thrive, jitteriness, and drug-induced parkinsonism (tremor and abnormal muscle movements) have been reported

The Art of Psychopharmacology

Potential Advantages

- Some cases of psychosis and bipolar disorder refractory to treatment with other antipsychotics
- * Often a preferred treatment for dementia with aggressive features
- * Often a preferred atypical antipsychotic for children with behavioral disturbances of multiple causations
- * Noncompliant patients (LAI risperidone)
- * Long-term outcomes may be enhanced when compliance is enhanced (LAI risperidone)

Potential Disadvantages

- Patients for whom elevated prolactin may not be desired (e.g., possibly pregnant patients; pubescent girls with amenorrhea; postmenopausal women with low estrogen who do not take estrogen replacement therapy)

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive functioning
- Unstable mood (both depression and mania)

- Aggressive symptoms

Pearls

* Well accepted for treatment of behavioral symptoms in children and adolescents, but may have more sedation and weight gain in pediatric populations than in adult populations

- Well accepted for treatment of agitation and aggression in elderly demented patients; however, numerous treatments are in late-stage clinical development for agitation and psychosis in dementia and will likely become first-line treatments rather than risperidone if approved
- Many anecdotal reports of utility in treatment-refractory cases and for positive symptoms of psychosis in disorders other than schizophrenia
- Hyperprolactinemia in women with low estrogen may accelerate osteoporosis
- Less weight gain than some antipsychotics, more than others
- Less sedation than some antipsychotics, more than others
- Increased risk of stroke may be most relevant in the elderly with atrial fibrillation
- May cause more motor side effects than some other atypical antipsychotics, especially when administered to patients with Parkinson's disease or Lewy body dementia

- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma drug levels and, if low, a dosage increase even beyond the usual prescribing limits
- Minimum therapeutic level is 28 ng/mL, with point of futility of 112 ng/mL (of active moiety, i.e., risperidone plus paliperidone)
- Patients with inadequate responses to atypical antipsychotics may also benefit from a trial of augmentation with a conventional antipsychotic or switching to a conventional antipsychotic
- However, long-term polypharmacy with a combination of a conventional antipsychotic with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Long-Acting Injectables

Consta	
Vehicle	Water
Tmax	21 days
T1/2 with multiple dosing	3–6 days
Time to reach steady state	6 weeks (4 injections)
Able to be loaded	No
Dosing schedule (maintenance)	2 weeks
Injection site	Intramuscular
Needle gauge	20 or 21
Dosage forms	12.5 mg, 25 mg, 37.5 mg, 50 mg
Injection volume	2 mL

Perseris	
Tmax	2 peaks: first at 4–6 hours and second at 10–14 days
Time to reach steady state	2 months (2 injections)
Able to be loaded	No (oral supplementation not needed)
Dosing schedule	1 month
Injection site	Abdomen (subcutaneous)

Perseris	
Needle gauge	18
Dosage forms	90 mg, 120 mg

Usual Dosage Range

- Consta: 12.5–50 mg every 2 weeks (deep intramuscular)
- Perseris: 90 or 120 mg every month (subcutaneous)

How to Dose

- Not recommended for patients who have not first demonstrated tolerability to oral risperidone (in clinical trials, 2 oral or short-acting IM doses are generally used to establish tolerability)
- Conversion from oral to Consta: oral coverage is required for 3–4 weeks; 2 mg oral risperidone is approximately 25 mg LAI every 2 weeks
- Conversion from oral to Perseris: no loading or oral supplementation; dose may be initiated at 90 mg (corresponding to 3 mg oral) or 120 mg (corresponding to 4 mg oral); administered once a month by subcutaneous injection in the abdomen

Dosing Tips –Depot

- With LAIs, the absorption rate constant is slower than the elimination rate constant, thus resulting in “flip-flop” kinetics; i.e.,